SECTION 1
THE IDEA AND THE TICKET JOURNEY

1.1 SYSTEM FUNCTION.

Private Pass is a test system for event entry. A person can give a copy of a ticket to someone else. Private Pass compares a person's face with the face data in their pass. A prepared gate can do this check without the internet. The design does not keep face data on the server for use across different events.

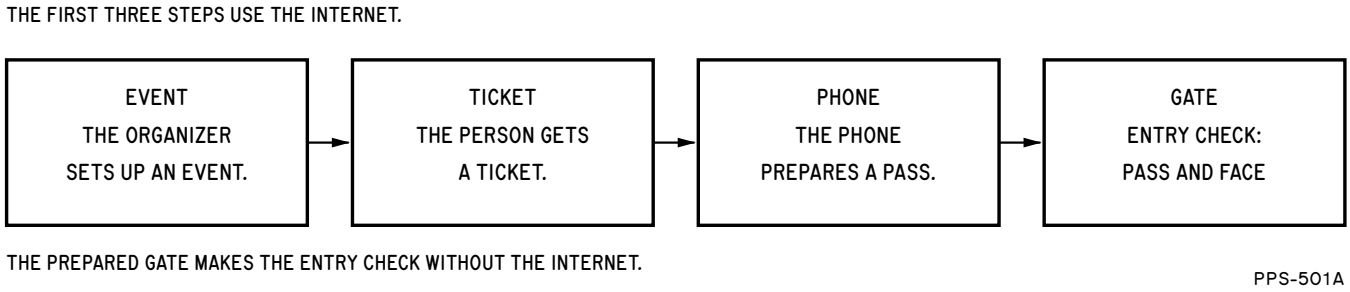


Figure 1. The Ticket Journey

PPS

1.2 THE FACE CHECK.

The phone changes a face image into numbers. These numbers are the face data. The phone changes these numbers for each event. The pass contains encrypted face data. The gate compares this data with the face at the entrance (figure 2).



Figure 2. One Person and One Pass

1.3 PASS PROTECTION.

The gate uses a digital signature to examine the source of a pass. The signature also shows changes to the pass. Encryption changes face data into a form that only a prepared gate can read. The gate must also compare the person's face with the pass.

Table 1. Words Used in This Handout

Term	Meaning
Pass	A digital ticket with encrypted face data.
Gate	The device that does the entry check.
Offline	Without an internet connection.
Server	The computer that keeps ticket records.
Prototype	A test system before use at real events.
Entry record	A saved record of ticket use.

NOTE

The server keeps accounts, events, and ticket records. This design does not remove all privacy risks.

KEY POINT

One person.
One pass.
A local entry check.

SECTION 2
TEST RESULTS AND LIMITS

2.1 WHAT THE TESTS SHOW.

The presentation gives results for software tests. These tests examine ticket rules, repeat requests, pass signatures, ticket reuse, and entry records. They do not show if the face check operates correctly at a real event.

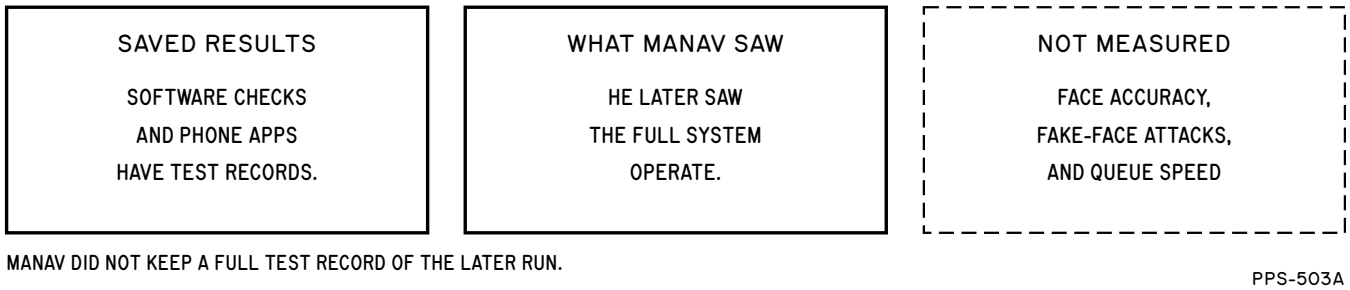


Figure 3. Three Types of Evidence

2.2 THE ENTRY CHECK.

The gate examines the event, time, digital signature, saved cancellations, previous use, and face. It saves ticket use before it shows an entry approval. When the gate connects to the internet again, it sends the entry record to the server.

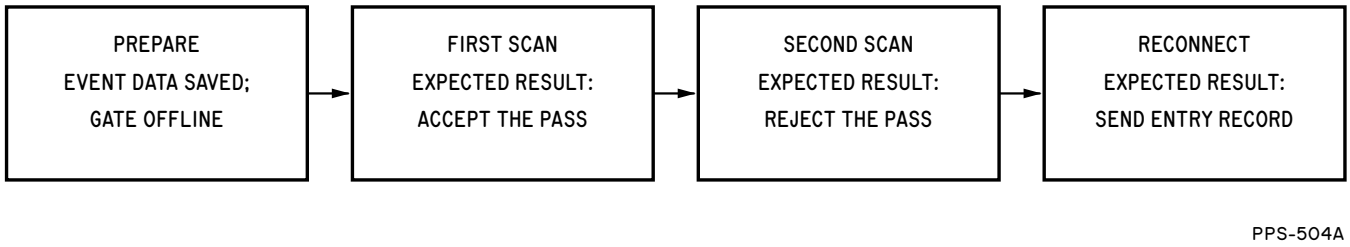


Figure 4. Demonstration: Expected Results

NOTE

Figure 4 shows expected results. The demonstration can give different results. An offline gate cannot receive new ticket cancellations.

2.3 MORE TESTS BEFORE REAL EVENTS.

More tests must measure correct and incorrect face matches. Tests must also examine attacks with photos, screens, and masks. Other tests must examine queue speed, different users, and entry without a face check.

2.4 THE PROJECT LESSON.

Manav added a website for tickets. This left less time for tests. The next project must have more time for tests and test records.

THE CONCLUSION.

Private Pass can operate in a prepared test system. Tests do not show that it can operate correctly at real events.

DISCUSSION	AUDIENCE
1 Which tests have records?	
2 What are the privacy risks?	
3 What is the next test?	

Source: PrivatePass_EPQ Presentation_v4.pptx and its speaker notes. Technical details are simplified for this handout.